

DL for ENG

LECTURE 9.1

Laminar Flow

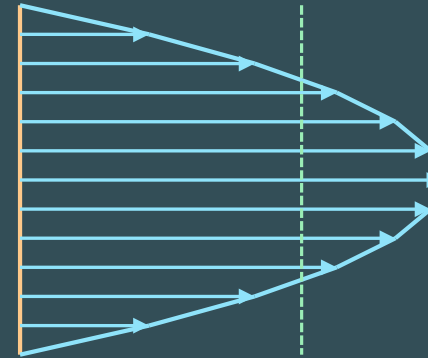
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

NEWTON · THE PIPE · DARCY



fastest on the axis

the mean: half the
centre speed

zero on the wall

Navier–Stokes is Newton's law for a parcel of liquid. In a straight tube far from its bends it reduces to one equation, solved by hand: Hagen–Poiseuille's parabola, Darcy's law, and the start-up when the pump begins. FDM and a PINN then solve it against the truth, and a flow meter gives the viscosity.

Learning Objectives

SECTIONS, IN ORDER

- 1 Flow equations** — Newton's law for a liquid, the Reynolds number
- 2 Pipe flow** — fully developed, Hagen–Poiseuille and Darcy, the start-up
- 3 Solving a flow** — conditions built in, FDM or PINN, the viscosity

White carries the pipe flow and Liu the method; the example is the exercise's own, the coolant in a 6 mm tube of a cold plate.

READ, AND WHAT EACH ANSWERS

White ch. 4 what does Newton say for a liquid?

White ch. 6 what is the flow in a tube?

Liu ch. 2.2 how is it solved, and checked?

Recap of L8.2 Transient Heat

1 - 2 · TIME

- 1** What does the storage term add to steady conduction, and what do the heat sink's three time scales say?

A term $\rho c_p \partial T / \partial t$, and so one initial field. Heat crosses the 10 mm base in 1.5 s, spreads along it in 15 s, and the whole sink warms and cools with the lumped time constant of 75 s: a 20 s pulse reaches only a quarter of the steady rise. *Incropera ch. 5; Ex_08 section 9*

- 2** How is causality put back into the PINN of the pulse, and where in time was its error largest?

The trial function is zero at the start and carries the lumped model: the hard initial condition, the first remedy. A square-root clock and half the points in the first 30 s sample early times densely. Its error was largest while the power was on. *Liu ch. 4.2 - 4.4; Ex_08 section 11*

3 - 4 · THE PULSE AND INVERSE

- 3** How was the transient checked, and why does the thermocouple peak after the power is off?

Against the exact transient at six instants: Crank–Nicolson meets 0.05 K on 321×65 nodes and 1200 steps, first order where the flux switches. The thermocouple sits 15 mm beside the module, and the heat still arriving keeps it rising for 3.5 s. *Incropera ch. 5; Ex_08 sections 9 - 11*

- 4** What was trained besides the weights in the inverse problem, and which part of the record identified it?

One number, the fins' Biot number as an exponential, entering the convection, the lumped model and the balance. The rise is set by the known heat capacity, the cooling tail after 20 s by h_{eff} alone, which was found 5 % low. *Liu ch. 4.6; Ex_08 section 12*

Navier–Stokes

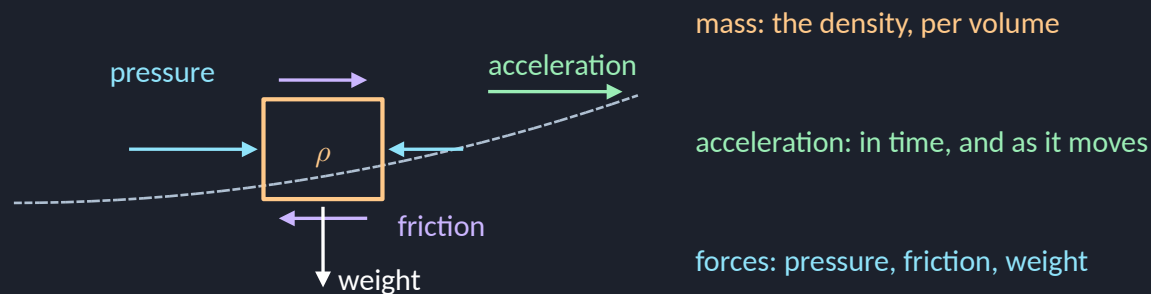
Navier–Stokes — Newton's second law for a small parcel of liquid: mass times acceleration equals the forces on it. The mass per volume is the density ρ ; the acceleration is how the parcel's velocity changes as it moves; the forces are the pressure, the friction of the neighbours, the weight.

MASS TIMES ACCELERATION EQUALS THE FORCES

$$\underbrace{\rho}_{\text{mass}} \underbrace{(\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u})}_{\text{acceleration}} = \underbrace{-\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}}_{\text{the forces}}, \quad \nabla \cdot \mathbf{u} = 0$$

ρ the mass per volume, $\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u}$ the acceleration of the parcel: in time, and because it moves on $-\nabla p$ the push of the pressure, $\mu \nabla^2 \mathbf{u}$ the friction of the neighbours ($\mu = \rho \nu$), $\rho \mathbf{g}$ the weight, $\nabla \cdot \mathbf{u} = 0$ the volume kept

A PARCEL OF LIQUID, AND NEWTON'S LAW ON IT



NEWTON'S LAW, TERM BY TERM

Newton	for a parcel of liquid	the term
mass	the density, per volume	ρ
acceleration	the velocity changes in time	$\mathbf{u}_t$
	and because it moves on	$(\mathbf{u} \cdot \nabla)\mathbf{u}$
force	the pressure, from the sides	$-\nabla p$
	the neighbours drag on it	$\mu \nabla^2 \mathbf{u}$
	its weight	$\rho \mathbf{g}$

Why $(\mathbf{u} \cdot \nabla)\mathbf{u}$ is an acceleration: in a time dt the parcel moves from x to $x + \mathbf{u} \cdot dt$, where the velocity is different, so its velocity changes by $\mathbf{u}_t \cdot dt + (\mathbf{u} \cdot dt \cdot \nabla)\mathbf{u}$. In a steady flow through a narrowing pipe $\mathbf{u}_t = 0$, yet every parcel speeds up.

Inertia and Darcy

Inertia term — $(\mathbf{u} \cdot \nabla)\mathbf{u}$, the one term where the velocity multiplies itself: the fluid carries its own momentum. Double $\mathbf{u}$ and every other term doubles, this one quadruples. Solutions can no longer be added, large eddies feed smaller ones, and the flow can turn turbulent.

FRAME IT, NEGLECT IT, AVERAGE IT

$$\rho \mathbf{u}_t + \boxed{\rho (\mathbf{u} \cdot \nabla)\mathbf{u}} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

keep it: nonlinear, turbulence

$$\rho \mathbf{u}_t = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

neglect it: Stokes, linear, slow flow

$$\bar{u} = -\frac{K}{\mu} \frac{dp}{dx}, \quad K = \frac{h^2}{12}$$

average across the channel: Darcy
the law this lecture rests on

$(\mathbf{u} \cdot \nabla)\mathbf{u}$ the inertia, the one nonlinear term; $\bar{u}$ the mean speed across the channel, dp/dx the pressure drop per length K the permeability, $h^2/12$ for a channel of height h , μ the viscosity, $\rho \mathbf{g}$ the weight

WHY IT IS NONLINEAR: DOUBLE THE VELOCITY

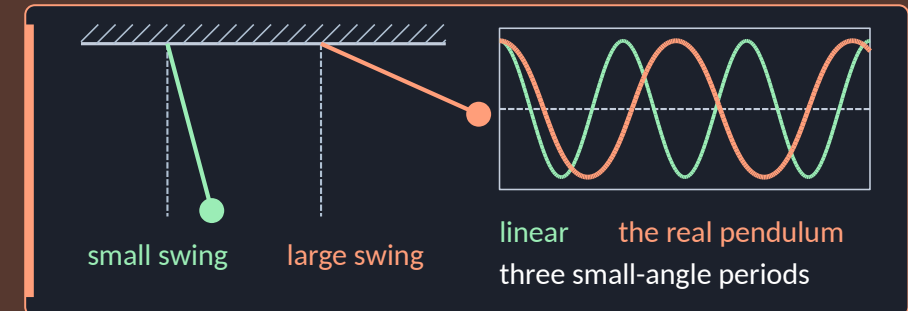
$\rho \mathbf{u}_t, \mu \nabla^2 \mathbf{u}, \nabla p$ double: linear

$\rho (\mathbf{u} \cdot \nabla)\mathbf{u}$ quadruples: $u \cdot u_x = (u^2/2)_x$

THE SAME STEP ON A PENDULUM

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0 \quad \Rightarrow \quad \ddot{\theta} + \frac{g}{\ell} \theta = 0$$

θ the angle, ℓ the length, g gravity; $\sin \theta \rightarrow \theta$ is the same step as dropping the inertia



Small swing, $\sin \theta \approx \theta$: linear, the same period at any amplitude - that is Stokes. Large swing: the period grows with the amplitude, driven it goes chaotic - that is Navier-Stokes. The amplitude decides whether the nonlinearity matters, as Re does for a flow.

The Navier–Stokes Millennium Problem

Millennium problem — given a smooth initial velocity field of finite energy, do the three-dimensional incompressible equations keep a smooth solution for all time, or can the speed become unbounded in finite time? One of the seven Clay Millennium Prize problems, stated so that a smooth applied force is allowed.

EULER FIRST, THEN NAVIER-STOKES

$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{f} \quad (\nu = 0)$$

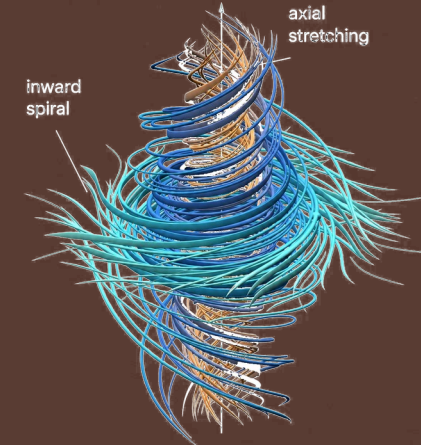
$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

$\mathbf{u}$ velocity, p pressure, ν viscosity, $\mathbf{f}$ an applied force; Euler is Navier-Stokes with no friction

THE ORDER OF THE MATHEMATICS, AND WHAT IS OPEN

- **Euler** — no friction: the easier case, broken first
- **Navier-Stokes with a force** — friction lost, machine-checked
- **open** — the unforced question; prize unclaimed

THE FLOW, AND WHO PROVED WHAT



the fluid spirals inward and is stretched along the axis; a thinner, faster core at every step, so the speed can grow without bound

who	what they proved
Martínez-Zoroa, Córdoba	the cascade, by hand
Alpöge, Buckmaster	forced Euler, announced first
OpenAI's system	unforced Euler, then forced N-S
Clay Institute	has not accepted it

The Reynolds Number

Reynolds number — the ratio of inertia to viscosity, $Re = \rho \bar{u} D / \mu$. In a pipe below about 2300 the flow stays laminar, layer sliding over layer; above it, it turns turbulent. It also sets how far the flow needs after a bend to settle into its profile: the entrance length, $0.05 Re D$.

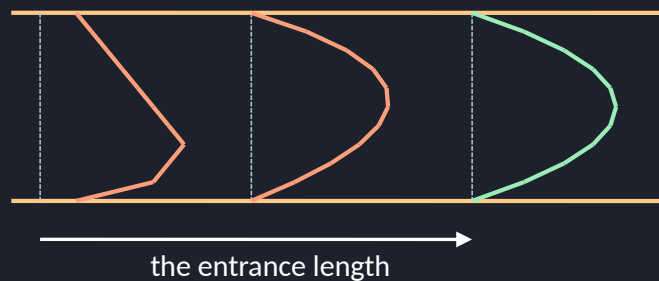
ONE NUMBER, AND THE LENGTH IT SETS

$$Re = \frac{\rho \bar{u} D}{\mu} = 452$$

$$L_e \approx 0.05 Re D = 136 \text{ mm}$$

ρ the density, $\bar{u}$ the mean speed, D the diameter, μ the viscosity
 L_e the entrance length: how far after a bend the profile needs to settle

AFTER A BEND, THE PROFILE SETTLES

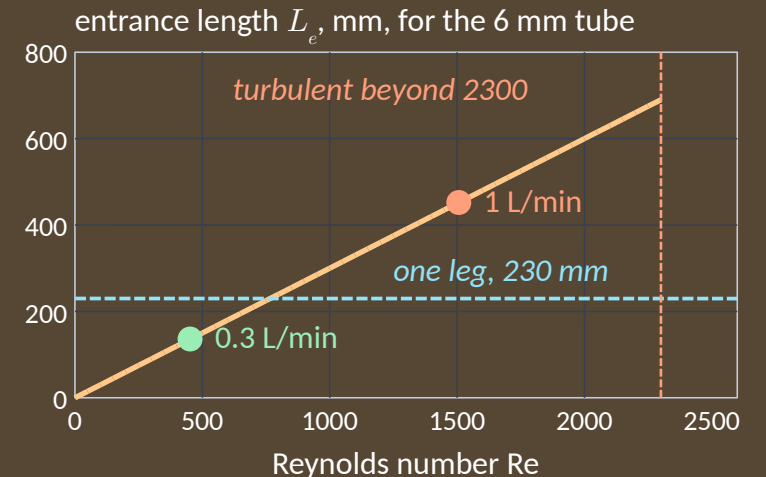


after a bend: skewed

settling along the straight leg

the parabola, fully developed

ENTRANCE LENGTH AGAINST Re



A leg is one straight run of the serpentine between two bends, 230 mm here. At 0.3 L/min the parabola forms 136 mm after each bend, inside the leg; at 1 L/min it would need 452 mm and never form.

Flow Boundary Conditions

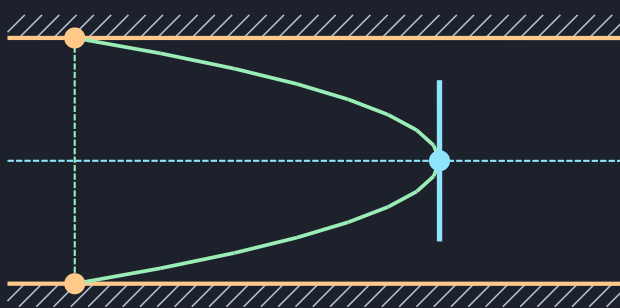
Flow boundary conditions — no-slip at a wall at rest: the fluid touching it is at rest too, $u = 0$ - an assumption, confirmed by experiment. In a tube $u = 0$ at $r = R$; on the axis the profile is symmetric, so $u_r = 0$; at the start the coolant is at rest; the pump sets the pressure gradient $G = -\partial p / \partial x$.

WALL, AXIS, START, PUMP

$$u(R, t) = 0, \quad u_r(0, t) = 0, \quad u(r, 0) = 0$$

u the velocity along the tube, r the distance from the axis, R the radius
 u_r its slope across: zero on the axis, where the profile is symmetric

A VALUE AT THE WALL, A SLOPE ON THE AXIS

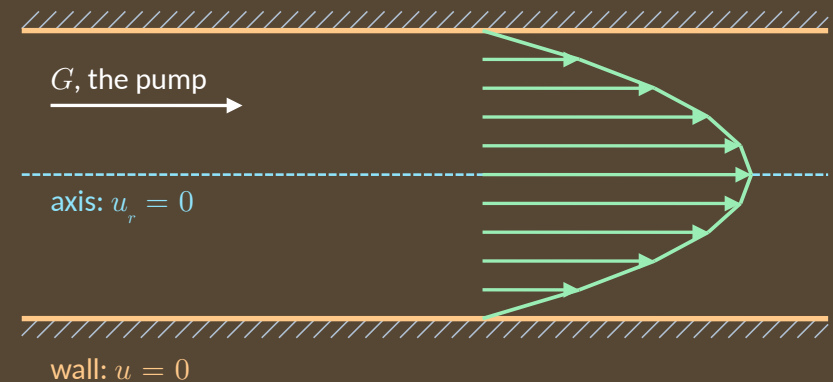


wall: value given (Dirichlet) - easy,
built into the trial function

axis: slope given (Neumann) -
harder: a loss term, or symmetry
built in

so the parabola crosses the axis flat

THE TUBE OF EX_09.1, ALONG ITS AXIS



wall $u = 0$: no slip, stuck not slowed

axis $u_r = 0$: symmetry

start $u = 0$: at rest

pump G held from $t = 0$

Fully Developed Flow

Fully developed flow — the flow in a straight tube far from its bends, where the profile no longer changes along the tube. Keeping the volume then forces the velocity towards the wall to zero, the inertia term is exactly zero, and only $u(r, t)$ along the tube is left: Stokes is the full equation.

STOKES ALONG THE TUBE, IN THREE LINES

$$\rho \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \nabla^2 u, \quad \frac{\partial p}{\partial x} = -G$$

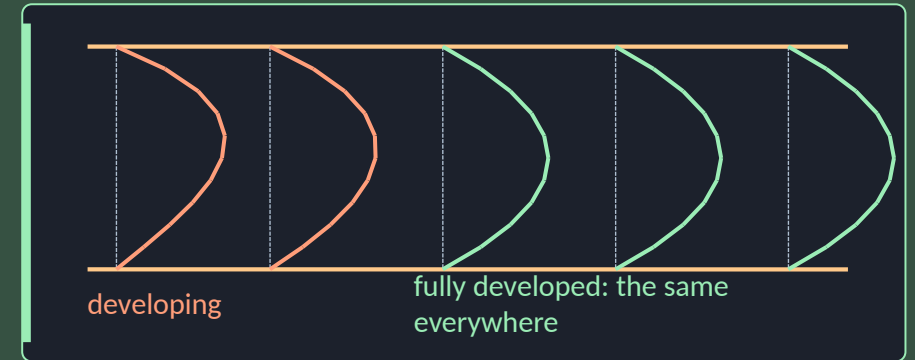
$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$\rho \frac{\partial u}{\partial t} = G + \mu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

u the velocity along the tube, x along, r from the axis, θ round it, ∂ a partial derivative
 G the pressure gradient, the drop per metre (Pa/m), μ the viscosity, ρ the density

Line 1, the x part of Stokes: p falls evenly along a straight tube, by G per metre, so $-\partial p / \partial x = G$. Line 2, the Laplacian in tube coordinates: the first term is zero (fully developed), the last is zero (round: the same at every angle θ). Line 3, what is left.

DEVELOPING, THEN DEVELOPED



round, straight

the same at every θ : no swirl

far from bends

$\partial u / \partial x = 0$: fully developed

volume kept

then $v = 0$: only $u(r, t)$

Hagen–Poiseuille Flow

Hagen–Poiseuille flow — the steady flow in a fully developed tube: a parabola, fastest on the axis and zero on the wall. Its mean speed is Darcy's law with the permeability of a round tube, $K = R^2/8$, exact here because the inertia is zero; written as a friction factor, $f = 64/Re$.

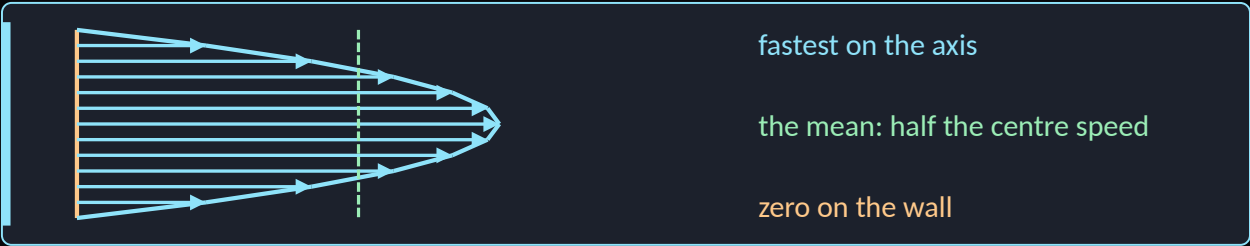
THE PARABOLA, ITS MEAN, ITS FRICTION

$$u(r) = \frac{GR^2}{4\mu} \left(1 - \frac{r^2}{R^2} \right)$$

$$f = \frac{64}{Re}$$

$$\bar{u} = \frac{GR^2}{8\mu} = \frac{K}{\mu} G, \quad K = \frac{R^2}{8}$$

R the tube's radius, G the pressure gradient (Pa/m), $\bar{u}$ the mean speed, K the permeability of a round tube
 f Darcy's friction factor, Re the Reynolds number



THE COLD PLATE'S TUBE, STEADY

quantity	value
mean speed $\bar{u}$	0.177 m/s
centre speed	0.354 m/s, twice the mean
pressure gradient G	393 Pa per metre
friction $f = 64/Re$	0.142
permeability K	1.1 mm ²

Why a parabola: the pressure pushes every shell alike. The friction between shells carries the push of everything inside out to the wall, so the shear stress grows with r , $\tau = Gr/2$, and u falls from the axis to zero at the sticking wall. Without friction every parcel would go alike.

Start-Up Flow

Start-up flow — the same tube from rest, when the pump starts. Its exact solution is the parabola less a sum of terms that each die away at their own rate. Hagen-Poiseuille is not a second answer: it is where the start-up ends, after about $R^2/\nu = 3.8$ s.

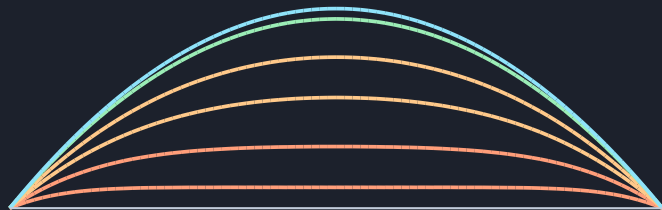
THE PARABOLA, LESS WHAT HAS NOT ARRIVED

$$u(r, t) = u(r) - \frac{GR^2}{4\mu} \sum_n c_n J_0\left(\frac{\lambda_n r}{R}\right) e^{-\lambda_n^2 \nu t / R^2}$$

$$c_n = \frac{8}{\lambda_n^3 J_1(\lambda_n)}, \quad J_0(\lambda_n) = 0$$

J_0, J_1 Bessel functions, λ_n the zeros of J_0 : 2.405, 5.520, 8.654

$\nu = \mu/\rho$ the kinematic viscosity, $u(r)$ the parabola, c_n the weight of each term

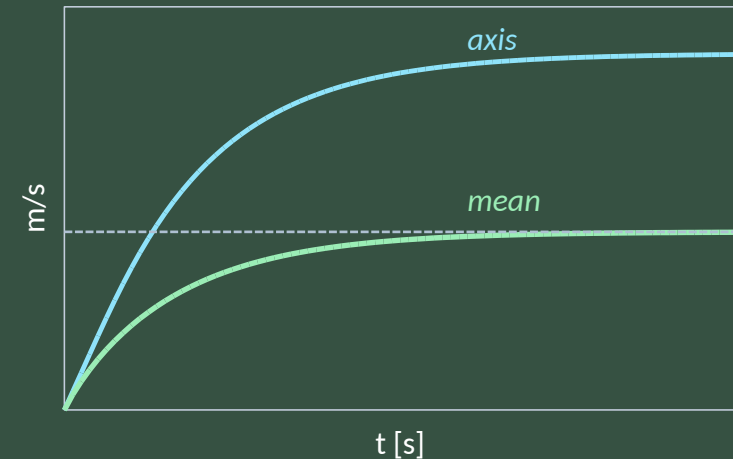


0.1 s: a flat core, pushed at once

1 s: rounding

4 s: the parabola

THE MEAN SPEED, AND THE AXIS



The mean speed - what a flow meter reads - is half-way after 0.43 s and within a per cent after about three. Ex_09.1's exact solution.

The Trial Function

Trial function — the network does not output the velocity; it outputs a free factor $\mathcal{N}$, and the velocity is built as $\hat{u} = (1 - s) \cdot \hat{t} \cdot \mathcal{N}$. The factor $1 - s$ is zero on the wall, $\hat{t}$ is zero at the start, so both conditions hold whatever $\mathcal{N}$ learns. The loss is then the equation alone.

WALL AND START BUILT IN: EX_09.1 TODO 1, 2

$$\hat{u}_{\hat{t}} = a + 4\kappa(s \hat{u}_{ss} + \hat{u}_s)$$

$$\hat{u} = (1 - s) \hat{t} \mathcal{N}(s, \hat{t})$$

$s = (r/R)^2$ where across the tube (0 axis, 1 wall), $\hat{t} = t/4s$ when (0 pump start, 1 window end)

$\hat{u} = u/u_c$ the velocity over the steady centre speed, $\mathcal{N}$ the network, κ the viscosity and a the push, scaled

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4)      # s and t in
u = (1 - s) * t * net(torch.cat([s, t], 1))      # wall and start, built in
us = grad(u, s); ut = grad(u, t)                # derivatives by autograd
r = (ut - a) / kappa - 4 * (s * grad(us, s) + us) # the residual, the whole loss
```

INPUTS, AND THE TWO FACTORS



$$s = (r/R)^2$$

where: 0 on the axis, 1 at the wall

$$\hat{t} = t / 4s$$

when: 0 pump start, 1 window end

$$1 - s$$

zero on the wall: no slip

$$\hat{t}$$

zero at the start: at rest

s is the squared distance from the axis, so it is the same on both sides of it: the symmetry is in the input.

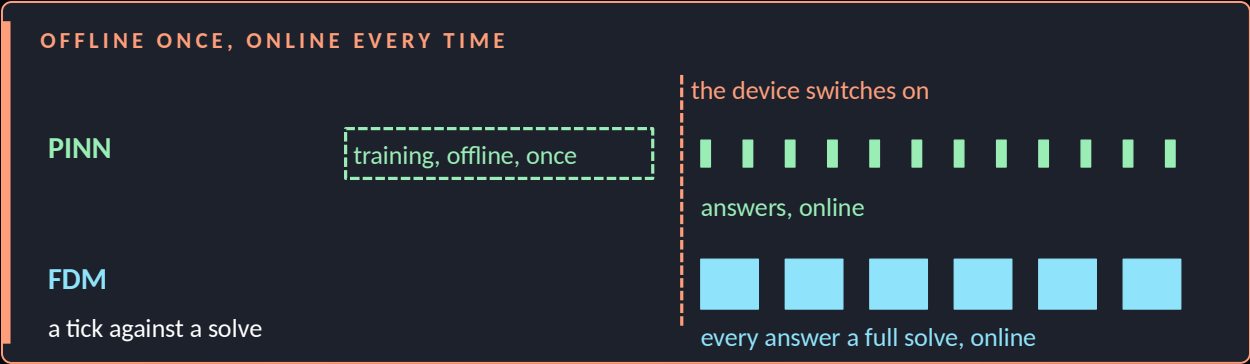
FDM or PINN

FDM or PINN — finite differences (FDM) compute an answer when asked. A PINN is trained before the device runs, offline; online, one evaluation is its answer. Compare online against online: one FDM solve against one evaluation, same device, same accuracy. Training is reported apart.

ONLINE AGAINST ONLINE, AT THE SAME ACCURACY

$S = \frac{t_{FDM}}{t_{PINN}}$ at the same error, on the same device

t_{FDM} one finite-difference solve, t_{PINN} one evaluation of the trained network; the training is offline and not in S



EX_09.1 SECTION 5, ON COLAB'S T4

measured	FDM	PINN
online, one answer	one solve, 41×160	one evaluation
accuracy	1e-03, the agreed	scored the same way
device	the T4, synchronised	the T4, synchronised
offline	none	the training, apart
grows with	nodes $\times$ steps per d	the points, slowly

Run section 5 on Colab's T4 and read the numbers off its bar chart: they are the ones to quote. On a problem this small the gap is tens, not thousands; the network's case is many answers at once, and the scaling test's extra dimensions.



Viscosity From a Flow Meter

Inverse problem — the coolant's viscosity μ is not known - it changes threefold with temperature and glycol - but a flow meter's record is. Any model that predicts the readings can be fitted to them: the exact series, FDM inside an optimiser, or a PINN with μ trained beside its weights.

WHERE THE VISCOSITY SITS IN THE LOSS

$$\mathcal{L} = \left[\frac{\hat{u}_i - a}{\kappa} - 4(s\hat{u}_{ss} + \hat{u}_s) \right]^2 + w_d \sum_k \left(\int_0^1 \hat{u} ds - \bar{u}_k \right)^2, \quad \kappa = e^\beta, \quad \mu = \kappa \frac{\rho R^2}{T}$$

κ the scaled viscosity inside the residual, $\beta = \ln \kappa$ the number Adam updates (keeps $\kappa > 0$; start $\beta = 0$, the nominal 2.5 mPa s)
 $\mu = \kappa \rho R^2 / T$ the viscosity in Pa s ($T = 4$ s), $\bar{u}_k$ the flow meter's mean speed at t_k , w_d the readings' weight

THE INVERSE IN PYTORCH

```
beta = nn.Parameter(torch.tensor(0.0)) # beta = ln kappa; 0: nominal
kappa = beta.exp() # kappa > 0, always
loss = res(net, kappa).pow(2).mean() + w_d * misfit # kappa divides res
opt = Adam([*net.parameters(), beta]) # trained with the weights
mu = kappa * rho * R**2 / T_END # back to Pa s
```

FOUR WAYS, ONE RECORD

method (true μ 3.50)	μ [mPa·s]	time to the answer
Darcy on the last reading	4.04	none
the exact series, fitted	3.55	instant
FDM inside an optimiser	3.53	5 solves, 12 ms
the PINN, μ trained	3.56	46 s of training

Ex_09.1, the flow meter's first second. Darcy reads the last value as steady and is 15 % high, because the flow had not settled. Any model that predicts the readings can be fitted; the network is one of them, and it pays where no cheap solver exists.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
fronts, Reynolds ceiling	a thin feature against the points	Liu ch. 5.6	mixing layers, high Re
stream function	continuity built in exactly	Liu ch. 2.5.3	2-D flows past bodies
pressure reference	the level the equations leave free	Liu ch. 2.2	any closed duct
bends: Dean flow	the vortices of a curved tube	White ch. 6	serpentines, MP9.1A
<u>porous media</u>	Darcy flow through a solid	Liu ch. 2.4	groundwater, CO ₂ storage
turbulent pipe flow	an averaged, modelled flow	Pope ch. 7	Re above 2300

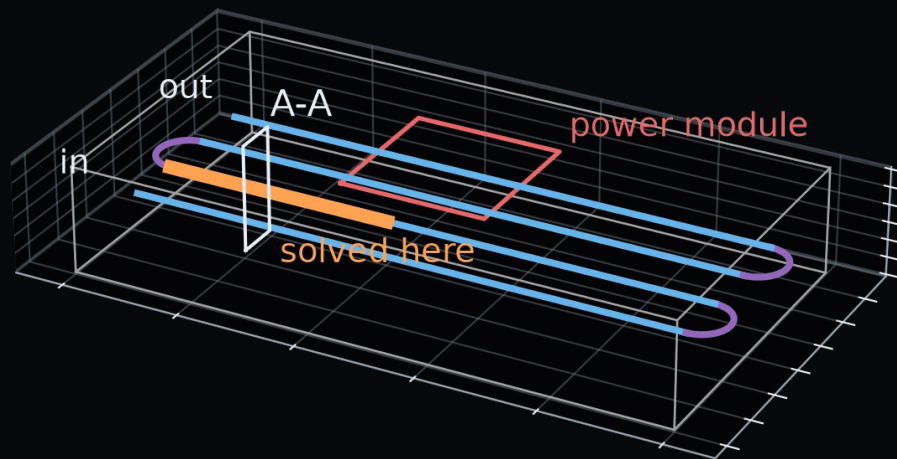
WHY THEY ARE LEFT OUT

Ex_09.1's tube is straight, laminar and fully developed, with an exact solution to score against, so none of them is needed to do it. The fourth row is mini project MP9.1A, where the students build the solver for a bend themselves.

Exercise 9.1

ONE NOTEBOOK, SECTION BY SECTION

- 1 - 3 — the cold plate, the physics, the exact start-up
- 4 — the exact solution, and finite differences
- 5 — a PINN, conditions built in; timed at equal accuracy
- 6 — the coolant's viscosity from a flow meter
- 7 - 9 — the report; mini projects MP9.1A and MP9.1B



THE QUESTION EACH PART ANSWERS

- 1 - 3 what is left of Navier-Stokes?
- 4 which grid is fine enough?
- 5 when does a PINN pay?
- 6 which way finds the viscosity?

REPORT QUESTION, TO CONCLUDE

At the same accuracy the trained PINN answered 70 times faster than FDM on one GPU, but only 5 times faster than SciPy on a CPU. Where in this cold plate would you choose the PINN, and what would you check first?

Key Takeaways

PIPE FLOW

what it is	Newton's law in a straight tube
function	the velocity across it, in time
train, tune	exact: parabola, Darcy, start-up
vs a bend	no inertia, no vortices
used for	pressure drop, pump, flow rate

SOLVING A FLOW

what it is	a PINN, conditions built in
function	$u(s, t)$, and κ from data
train, tune	the equation alone; then L-BFGS
vs FDM	slower once, faster after N^*
used for	many queries, many dimensions

Next — L9.2: a gas in a fuel-cell channel, and what it carries.

Questions

1 – 2 • PIPE FLOW

- 1 **What is left of Navier–Stokes in a straight tube far from its bends, and why?**
Only the velocity along the tube, changing across it and in time. Round and straight, nothing makes it swirl; far from the bends, nothing changes along the tube; keeping the volume then makes the velocity towards the wall zero, so the inertia term vanishes. *White ch. 6; Ex_09.1 section 3*

- 2 **What do Hagen–Poiseuille and Darcy's law say, and when do they hold?**
The steady profile is a parabola, and the mean speed is the pressure drop times $R^2/8\mu$: Darcy's law with $K = R^2/8$, or $f = 64/Re$. They hold for laminar flow, Re below about 2300, beyond the entrance length $0.05 Re D$ after a bend. *White ch. 6; Ex_09.1 section 3*

3 – 4 • SOLVING A FLOW

- 3 **Which conditions can be built into a PINN for the tube, and how?**
The wall and the start by factors in the trial function, $(1 - s) t$, and the symmetry about the axis by taking $s = r^2/R^2$ as the input. Whatever the weights, all three hold, so the loss is the equation alone. *Liu ch. 2.2; Ex_09.1 section 5*

- 4 **When is a PINN worth more than FDM, and how is a viscosity found from data?**
When many answers are wanted or FDM grows out of reach: a trained PINN answers fast but repays its training only after many runs. The viscosity is fitted to a flow meter's readings by the exact series, FDM in an optimiser or a network. *Ex_09.1 sections 5 – 6*

NEXT

L9.2 — Gas Flow

A gas in the fuel channel of a solid oxide cell: the flow is known, and the question is what it carries - hydrogen in, steam out - along the channel.

BEFORE THE NEXT SESSION

read	White ch. 6
run	Ex_09.1, sections 1 to 7
bring	your viscosity from the flow meter
think about	where you need many answers

*A laminar flow, solved exactly and by two methods.
Next, a flow that carries something along with it.*